

VALO V5.0 — Technical Whitepaper

Formally Verified AI Safety for Critical Infrastructure

Version: V5.0 — Validation

Phase

Date: May 15, 2026

Classification: Confidential — NDA Required

Validation Status: VALO V5.0 is in the validation phase. TLA+ specification is complete and ready for independent model checker verification. Empirical test suite is designed but not yet executed. UiS validation is the next step.

Abstract

VALO V5.0 is a three-layer deterministic AI safety engine for critical infrastructure under EU AI Act Annex III. L1-Guardian (Rust, 180 lines) implements a formally specified ACCEPT/REJECT gate. L2-Orchestrator handles logging and routing. L3-Context Engine manages source trust scoring. This whitepaper presents the specification for academic review.

1. Three-Layer Architecture

Layer	Component	Language	Function
L1	Guardian	Rust	Safety decision gate — ACCEPT/REJECT in 43ns. 0 unsafe blocks. Only formally specified layer.
L2	Orchestrator	Python	Logging, routing, authentication, SHA-256 WORM audit log. Cannot override L1.
L3	Context Engine	Rust/Python	Source reliability tracking, distrust escalation (L0 Trusted → L4 Total Halt).

Architectural guarantee: L2 and L3 cannot override L1's decision. If L1 says REJECT, the system halts. This is enforced architecturally, not by convention.

2. Formal Specification (TLA+)

The L1-Guardian is modelled as a state machine in TLA+. Specification is complete and ready for independent model checker (TLC) verification.

States & Transitions

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States: {COLD, WARMING, ACTIVE, ACCEPT, REJECT} Transitions: 6 defined Flow: COLD ->
WARMING -> ACTIVE -> ACCEPT (safe) -> REJECT (unsafe) -> COLD (rollback) Source:
formal-verification/ValoStateMachine.tla
```

Four Invariants — Specified, Pending Verification

IntegrityInvariant SPECIFIED — PENDING VERIFICATION

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(safe_flag = TRUE) => (output is guaranteed safe)
```

AtomicityInvariant SPECIFIED — PENDING VERIFICATION

(Execute OR Rollback) => (all state updates, or none)

DeterminismInvariant SPECIFIED — PENDING VERIFICATION

(same input AND same state) => (same output)

LivenessInvariant SPECIFIED — PENDING VERIFICATION

(request != NULL) => (responds or halts)

Temporal Properties

Deadlock Freedom: SPECIFIED

No Livelock: SPECIFIED

3. Implementation

L1-Guardian	Rust, 180 lines, 0 unsafe blocks, 43ns target. Source: l1-guardian/src/
L2-Orchestrator	Python, SHA-256 WORM log, TCP/UDS transport. Source: l2-orchestrator/
L3-Context Engine	Rust/Python, distrust matrix L0–L4. Source: l3-context/src/
Frame protocol	64-byte ValoFrame, CRC32C. Source: l1-guardian/src/valo_frame.rs
Security audit	May 2026 — 0 critical findings, 0 memory vulnerabilities

4. Honest Scope

Complete:

- ✓ TLA+ spec — 5 states, 6 transitions, 4 invariants, 2 temporal properties
- ✓ L1-Guardian Rust implementation — 0 unsafe blocks
- ✓ Security audit — 0 critical findings
- ✓ L3 distrust matrix — defined and implemented

Pending UiS validation:

- Model checker (TLC) has not yet run — this is what we ask UiS to execute
- Empirical test suite designed (1M cases) — not yet executed
- L2 and L3 are not formally specified — only L1 is in TLA+ scope

GitHub Repository Structure

Repository: Valo-Research-Geoup/valo-v5-core (private)

/	
VALIDATION.md	Formal validation specification and scope
WHITEPAPER.md	Technical whitepaper (Markdown)
SOVEREIGNTY.md	IP sovereignty and confidentiality declaration
Dockerfile	Container build definition
simulate.py	Infrastructure simulation runner
simulate_vaig.py	VAIG (AI inference guard) simulation
/l1-guardian/	
src/main.rs	L1 Guardian entry point — Rust

src/lib.rs	Core safety logic — ACCEPT/REJECT gate
src/valo_frame.rs	64-byte cache-aligned ValoFrame struct
src/validation_logic.rs	Safety measurement logic
src/crc32c.rs	CRC32C integrity verification
src/server.rs	Socket server — UDS/TCP transport
/l2-orchestrator/	
ai_gateway.py	AI model interface and routing
src/auth.py	Authentication layer
src/worm_log.py	SHA-256 hash-chained WORM audit log
src/propagator.py	Frame propagation and transport
/l3-context/	
src/context_engine.rs	Source reliability engine — Rust
src/distrust.rs	Distrust escalation matrix (L0–L4)
src/feeds.rs	External data feed management
/formal-verification/	
ValoStateMachine.tla	Primary TLA+ specification — L1 state machine
ValoStateMachine.cfg	TLC model checker configuration
valo_v5.tla	Extended TLA+ specification
/docs/	
Valo_Research_Group___WP_Valo_5_0_LOCKED.pdf	Locked whitepaper PDF
/sidecar/	
vaig.py	VAIG — AI inference guard sidecar
client.py	Test client